LEARNER GUIDE

Direct and inverse proportion

Percentages, Ratio, Growth and Bounds · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Two quantities are in **direct proportion** when $y = kx$: double x and y doubles, $y \div x$ is always k, and the graph is a straight line through the origin. They are in **inverse proportion** when $y = k/x$: double x and y halves, $x \times y$ is always k, and the graph is a curve that never meets the axes. k is the **constant of proportionality**. One pair of values is enough to find it.

What you are learning to notice

- Direct: $y \div x$ stays the same. Inverse: $x \times y$ stays the same.
- A straight line is direct proportion only if it passes through the origin.
- If $y \propto x^2$, doubling x multiplies y by 4.
- One pair of values fixes k. Then the formula gives every other value.
- The graph of $y = k/x$ never touches either axis.

Moving through it, one step at a time

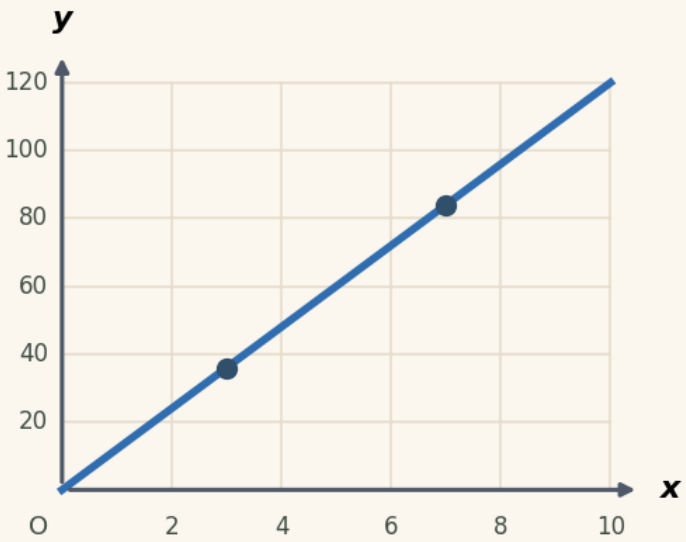
Decide the type (direct, inverse, square, root). Write the formula with k, for example $y = kx$. Substitute the pair you know to find k. Write the formula with the number in place of k, then substitute the new value.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Direct proportion: pay by the hour

Pay y (£) is directly proportional to hours worked x. 3 hours pay £36. How much do 7 hours pay?

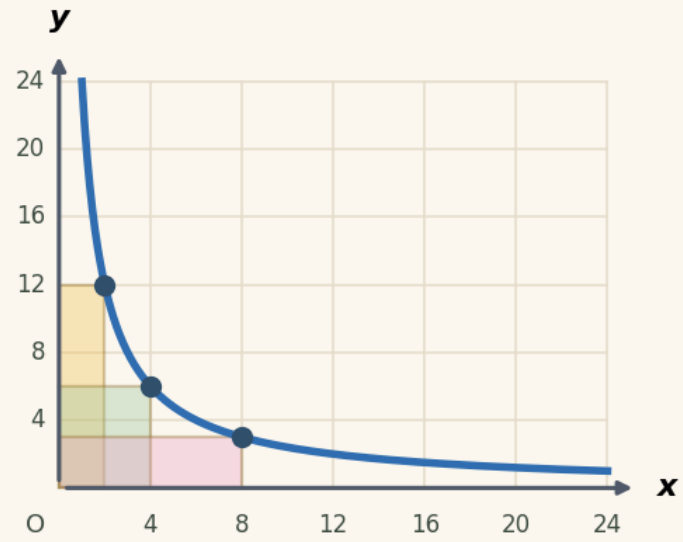


1. Direct proportion $y = kx$
2. Substitute $x = 3$, $y = 36$ $36 = 3k$, so $k = 12$
3. The formula $y = 12x$
4. $x = 7$ $y = 12 \times 7 = 84$

7 hours pay £84. The graph is a straight line through the origin.

Example 2 — Inverse proportion: sharing a job

The time each worker takes, y hours, is inversely proportional to the number of workers, x. 2 workers take 12 hours each. How long do 8 workers take?



1. Inverse proportion $y = \frac{k}{x}$, so $x \times y = k$
2. Substitute $x = 2$, $y = 12$ $k = 2 \times 12 = 24$
3. The formula $y = \frac{24}{x}$
4. $x = 8$ $y = \frac{24}{8} = 3$

8 workers take 3 hours each. Every shaded rectangle has area 24.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the unitary method (Lesson 1), substitution, and squares and square roots), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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